

Caleb James Smith

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Summary

Software developer and data scientist specialized in automation and data processing, visualization, and analysis. Ten years of experience in particle physics research as a collaborator on the Compact Muon Solenoid (CMS) experiment. Experienced, persistent, and creative problem solver. Hardworking team player. Curious learner, researcher, and teacher.

Technical Strengths

Languages	Python, C++, Bash, HTML, CSS
Libraries	Matplotlib, NumPy, Pandas, Flask, Django, ROOT
Tools	Unix/Linux Terminal, VS Code, Vim, Git, JSON, XML, Markdown, L ^A T _E X, Claude Code, Microsoft Office

Work Experience

AI Evaluator (Part Time) <i>Data Annotation</i>	Feb. 2026–Present <i>Remote</i>
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- Reviewed and graded the quality of AI responses (text output from LLMs). Created custom, realistic system and user prompts and evaluated model responses.

Postdoctoral Researcher (Physics) <i>The University of Kansas</i>	Jan. 2021–Jan. 2026 <i>Lawrence, KS</i>
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- Analyzed CMS data and Monte Carlo simulation (billions of events) using over 2,000 independent search regions in a data-driven fit to search for supersymmetric particles. Used ML algorithms (DNNs) to identify bottom quarks.
- Developed an automated electrical testing framework (Python) to test high-speed (1.28 Gbps) electrical cables (e-links). Reduced testing time by a factor of 10. Framework used to test over 1,000 e-links for the CMS pixel tracker upgrade.
- Created a visualization of e-link production for each stage over time (Python, Pandas, NumPy, Matplotlib). Demonstrated that e-link production was projected to finish 1.5 years ahead of schedule.

Research Assistant (Physics) <i>Baylor University</i>	May 2016–Dec. 2020 <i>Fermilab (Batavia, IL) and CERN (Meyrin, Switzerland)</i>
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- Analyzed CMS data and Monte Carlo simulation (billions of events) to search for supersymmetric particles. Used data-driven analysis to predict the Z to neutrinos background and associated uncertainties. Used ML algorithms (DNNs) to identify top/bottom quarks and W bosons.
- Created and maintained software with a team at Fermilab, including GUI, database, and website (Python, Tkinter, Django, Bash), to test over 700 electronic readout cards for the CMS hadron calorimeter upgrade.

Teaching Assistant (Physics), Instructor of Record <i>Baylor University</i>	Aug. 2014–May 2016 <i>Waco, TX</i>
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- Taught undergraduate lectures (Physics 2) and labs (Physics 1 and 2).

Personal Projects

Chess website	Interactive chess website for human/AI players built with HTML/CSS/JavaScript using Claude Code.
Speedcubing app	Rubik's Cube algorithm Android app built using Android Studio, Kotlin, and Jetpack Compose.

Education

Baylor University PhD in Physics Dissertation: <i>Search for supersymmetric top quarks in the CMS Run 2 data set</i>	Aug. 2014–Dec. 2020
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Baylor University MA in Physics	Aug. 2014–May 2018
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Taylor University BS in Physics, Minor in Mathematics	Aug. 2011–May 2014
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